

Grocery Price Analysis

SQL Queries — Statistics Canada Table 18-10-0245

Exploration & Filtering

Step 1: Row counts by province

Confirms the data is complete for each province. Whitehorse and Yellowknife will show far fewer records than the provinces — that's expected, since those two were added to the survey later.

```
SELECT
    GEO AS province,
    COUNT(*) AS number_of_records
FROM grocery_data
GROUP BY province
ORDER BY number_of_records DESC;
```

Step 2: Remove the 4 non-food items

Confirms how many grocery-only records remain once deodorant, laundry detergent, shampoo, and toothpaste are excluded from the dataset.

```
SELECT COUNT(*) AS grocery_only_record_count
FROM grocery_data
WHERE Products NOT IN (
    'Deodorant, 85 grams',
    'Laundry detergent, 4.43 litres',
    'Shampoo, 400 millilitres',
    'Toothpaste, 100 millilitres'
);
```

Step 3: One product across every province, right now

Shows what ground beef costs in every province as of the most recent month in the data (July 2026). Swap the product name to explore any other grocery item.

```
SELECT
    GEO AS province,
    Products AS grocery_item,
    VALUE AS current_price
FROM grocery_data
WHERE Products = 'Ground beef, per kilogram'
    AND REF_DATE = '2026-07'
ORDER BY current_price DESC;
```

Step 4: Which province has the highest grocery prices overall?

Averages every grocery item's price per province for the most recent month. Answers Business Problem #1 — where consumers are paying the most.

```
SELECT
    GEO AS province,
    ROUND(AVG(VALUE), 2) AS average_grocery_price
FROM grocery_data
WHERE REF_DATE = '2026-07'
    AND GEO != 'Canada'
GROUP BY province
ORDER BY average_grocery_price DESC;
```

Comparing Years

Step 5a: Average price per item, 2017

Establishes a baseline — the average price of every grocery item in the earliest full year of data.

```
SELECT
    Products AS grocery_item,
    ROUND(AVG(VALUE), 2) AS price_in_2017
FROM grocery_data
WHERE GEO = 'Canada'
    AND REF_DATE LIKE '2017%'
GROUP BY grocery_item
ORDER BY price_in_2017 DESC;
```

Step 5b: Average price per item, 2025

Run separately from Step 5a, then compare the two result tables side by side (e.g. in Excel or Google Sheets) to see which items rose the most in dollar terms. Together, these two steps answer Business Problem #2 (largest increases) and Business Problem #4 (which basket items drive rising costs most).

```
SELECT
    Products AS grocery_item,
    ROUND(AVG(VALUE), 2) AS price_in_2025
FROM grocery_data
WHERE GEO = 'Canada'
    AND REF_DATE LIKE '2025%'
GROUP BY grocery_item
ORDER BY price_in_2025 DESC;
```

Step 6: Which items swing in price the most (dollar terms)?

Finds each item's highest price ever minus its lowest price ever seen in the dataset. Answers Business Problem #3 — price instability.

```
SELECT
  Products AS grocery_item,
  ROUND(MIN(VALUE), 2) AS lowest_price_ever,
  ROUND(MAX(VALUE), 2) AS highest_price_ever,
  ROUND(MAX(VALUE) - MIN(VALUE), 2) AS price_swing
FROM grocery_data
WHERE GEO = 'Canada'
GROUP BY grocery_item
ORDER BY price_swing DESC
LIMIT 15;
```

Step 7: Which items swing the most relative to their own price?

Reframes Step 6 as a percentage instead of a raw dollar amount — a \$28 swing on a \$25 item is a much bigger story than a \$28 swing on a \$200 item. Answers Business Problem #5 — unusual price movements that may point to supply issues.

```
SELECT
  Products AS grocery_item,
  ROUND(AVG(VALUE), 2) AS average_price,
  ROUND(MAX(VALUE) - MIN(VALUE), 2) AS price_swing,
  ROUND(100.0 * (MAX(VALUE) - MIN(VALUE)) / AVG(VALUE), 1)
    AS swing_as_percent_of_price
FROM grocery_data
WHERE GEO = 'Canada'
GROUP BY grocery_item
ORDER BY swing_as_percent_of_price DESC
LIMIT 15;
```